Chaemin Lim

High Performance Computing Platform (HPCP) Lab.
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Research Interests

- Database Systems on Modern Hardware
- Concurrent Computing

Publications

- FaScalSQL: A Fast and Scalable GPU-Accelerated SQL Query Engine for Out-of-Memory Tables
  Chaemin Lim, Suhyun Lee, Jinwoo Choi, Kwanghyun Park, Jinho Lee, Joonsung Kim, and Youngsok Kim
  42nd IEEE International Conference on Data Engineering (**ICDE**), May 2026

- DMO-DB: Mitigating the Data Movement Bottlenecks of GPU-Accelerated Relational OLAP
  Chaemin Lim, Suhyun Lee, Jinwoo Choi, Joonsung Kim, Jinho Lee, and Youngsok Kim
  34th International Conference on Parallel Architectures and Compilation Techniques (**PACT**), Nov. 2025

- SPID-Join: A Skew-resistant Processing-in-DIMM Join Algorithm Exploiting the Bank- and Rank-level Parallelisms of DIMMs
  Suhyun Lee, **Chaemin Lim**, Jinwoo Choi, Heelim Choi, Chan Lee, Yongjun Park, Kwanghyun Park, Hanjun Kim, and Youngsok Kim
  2025 ACM International Conference on Management of Data (**SIGMOD**), June 2025

- PID-Comm: A Fast and Flexible Collective Communication Framework for Commodity Processing-in-DIMMs
  Si Ung Noh[1], Junguk Hong[1], **Chaemin Lim**, Seongyeon Park, Jeehyun Kim, Hanjun Kim, Youngsok Kim, and Jinho Lee
  51st IEEE/ACM International Symposium on Computer Architecture (**ISCA**), June–July 2024

- Design and Analysis of a Processing-in-DIMM Join Algorithm: A Case Study with UPMEM DIMMs
  Chaemin Lim, Suhyun Lee, Jinwoo Choi, Jounghoo Lee, Seongyeon Park, Hanjun Kim, Jinho Lee, and Youngsok Kim
  2023 ACM SIGMOD International Conference on Management of Data (**SIGMOD**), June 2023

- GuardiaNN: Fast and Secure On-Device Inference in TrustZone Using Embedded SRAM and Cryptographic Hardware
  Jinwoo Choi[1], Jaeyeon Kim[1], **Chaemin Lim**[1], Suhyun Lee, Jinho Lee, Dokyung Song, and Youngsok Kim
  23rd ACM/IFIP International Middleware Conference (**Middleware**), Nov. 2022

Projects

- **Development of New Concept PIM Semiconductor Leading Technology**
  Institute for Information and communication Technology Planning and evaluation (IITP) — *2022*
  - Presented UPMEM-based Processing-in-Memory (PIM) research at ABUMPIMP, a minisymposium on applications and benefits of commercial massively parallel PIM platforms.
  - Open-sourced research artifacts for PID-Join (SIGMOD 2023) and PID-Comm (ISCA 2024), enabling reproducible evaluation on commodity UPMEM DIMMs.
  - Developed PIM database and communication systems targeting data movement bottlenecks, join processing, and collective communication on modern memory-centric hardware.
- **STACK: Smart, Attack-Resistant Internet of Things Networks**
  Korea Institute for Advancement of Technology (KIAT) — *2021*
  - Reported an STM32MP1 clock-rate computation bug in OP-TEE OS affecting HASH1, GPIOZ, CRYP1, and MDMA clocks; acknowledged by a merged upstream fix with a `Reported-by` tag. OP-TEE/optee_os#4904
  - Worked with secure embedded-system software stacks, including trusted execution environment (TEE) components for attack-resistant IoT platforms.

Education

- **Yonsei University** Seoul, Korea
  Ph.D Student, Computer Science *September 2023 - Present*
- **Yonsei University** Seoul, Korea
  Master, Artificial Intelligence *March 2021 - August 2023*
- **Hankuk University of Foreign Studies** Seoul, Korea
  B.Eng. in Computer Engineering *March 2017 - February 2021*
- **Hankuk University of Foreign Studies** Seoul, Korea
  B.A. in Journalism *March 2017 - February 2021*

Teaching Experience

- **Logic Circuit Design (CSI2111)**: Teaching Assistant, Fall 2022
- **Open Source Projects (CSI2115)**: Teaching Assistant, Fall 2021
- **Information Security (CSI4109)**: Teaching Assistant, Spring 2021

Skills

- **C, C++, Python, Scala, Rust, SQL**